Assignment 1

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Q1

Install given packages

```
install.packages(c('plyr', 'MASS', 'ggplot2', 'dplyr'), repos =
↪  "https://cran.icts.res.in/")
```

```
## Installing packages into 'C:/Users/singh/AppData/Local/R/win-library/4.3'
## (as 'lib' is unspecified)

## package 'plyr' successfully unpacked and MD5 sums checked
## package 'MASS' successfully unpacked and MD5 sums checked
## package 'ggplot2' successfully unpacked and MD5 sums checked
## package 'dplyr' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
##  C:\Users\singh\AppData\Local\Temp\Rtmp4aI6Zu\downloaded_packages
```

Q2

Find answers to log2(2*5) and log(exp(1)exp(1))

```r
log2(2^5)
```

```
## [1] 5
```

```r
log(exp(1)*exp(1))
```

```
## [1] 2
```

Q3

Implement functions using built-in dataset

```r
data(iris) #loading the dataset
iris
```

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
## 1            5.1         3.5          1.4         0.2     setosa
## 2            4.9         3.0          1.4         0.2     setosa
## 3            4.7         3.2          1.3         0.2     setosa
## 4            4.6         3.1          1.5         0.2     setosa
## 5            5.0         3.6          1.4         0.2     setosa
## 6            5.4         3.9          1.7         0.4     setosa
## 7            4.6         3.4          1.4         0.3     setosa
## 8            5.0         3.4          1.5         0.2     setosa
## 9            4.4         2.9          1.4         0.2     setosa
## 10           4.9         3.1          1.5         0.1     setosa
## 11           5.4         3.7          1.5         0.2     setosa
## 12           4.8         3.4          1.6         0.2     setosa
## 13           4.8         3.0          1.4         0.1     setosa
## 14           4.3         3.0          1.1         0.1     setosa
## 15           5.8         4.0          1.2         0.2     setosa
## 16           5.7         4.4          1.5         0.4     setosa
## 17           5.4         3.9          1.3         0.4     setosa
## 18           5.1         3.5          1.4         0.3     setosa
## 19           5.7         3.8          1.7         0.3     setosa
## 20           5.1         3.8          1.5         0.3     setosa
## 21           5.4         3.4          1.7         0.2     setosa
```

```
## 22          5.1         3.7         1.5         0.4     setosa
## 23          4.6         3.6         1.0         0.2     setosa
## 24          5.1         3.3         1.7         0.5     setosa
## 25          4.8         3.4         1.9         0.2     setosa
## 26          5.0         3.0         1.6         0.2     setosa
## 27          5.0         3.4         1.6         0.4     setosa
## 28          5.2         3.5         1.5         0.2     setosa
## 29          5.2         3.4         1.4         0.2     setosa
## 30          4.7         3.2         1.6         0.2     setosa
## 31          4.8         3.1         1.6         0.2     setosa
## 32          5.4         3.4         1.5         0.4     setosa
## 33          5.2         4.1         1.5         0.1     setosa
## 34          5.5         4.2         1.4         0.2     setosa
## 35          4.9         3.1         1.5         0.2     setosa
## 36          5.0         3.2         1.2         0.2     setosa
## 37          5.5         3.5         1.3         0.2     setosa
## 38          4.9         3.6         1.4         0.1     setosa
## 39          4.4         3.0         1.3         0.2     setosa
## 40          5.1         3.4         1.5         0.2     setosa
## 41          5.0         3.5         1.3         0.3     setosa
## 42          4.5         2.3         1.3         0.3     setosa
## 43          4.4         3.2         1.3         0.2     setosa
## 44          5.0         3.5         1.6         0.6     setosa
## 45          5.1         3.8         1.9         0.4     setosa
## 46          4.8         3.0         1.4         0.3     setosa
## 47          5.1         3.8         1.6         0.2     setosa
## 48          4.6         3.2         1.4         0.2     setosa
## 49          5.3         3.7         1.5         0.2     setosa
```

```
## 50          5.0         3.3         1.4         0.2     setosa
## 51          7.0         3.2         4.7         1.4 versicolor
## 52          6.4         3.2         4.5         1.5 versicolor
## 53          6.9         3.1         4.9         1.5 versicolor
## 54          5.5         2.3         4.0         1.3 versicolor
## 55          6.5         2.8         4.6         1.5 versicolor
## 56          5.7         2.8         4.5         1.3 versicolor
## 57          6.3         3.3         4.7         1.6 versicolor
## 58          4.9         2.4         3.3         1.0 versicolor
## 59          6.6         2.9         4.6         1.3 versicolor
## 60          5.2         2.7         3.9         1.4 versicolor
## 61          5.0         2.0         3.5         1.0 versicolor
## 62          5.9         3.0         4.2         1.5 versicolor
## 63          6.0         2.2         4.0         1.0 versicolor
## 64          6.1         2.9         4.7         1.4 versicolor
## 65          5.6         2.9         3.6         1.3 versicolor
## 66          6.7         3.1         4.4         1.4 versicolor
## 67          5.6         3.0         4.5         1.5 versicolor
## 68          5.8         2.7         4.1         1.0 versicolor
## 69          6.2         2.2         4.5         1.5 versicolor
## 70          5.6         2.5         3.9         1.1 versicolor
## 71          5.9         3.2         4.8         1.8 versicolor
## 72          6.1         2.8         4.0         1.3 versicolor
## 73          6.3         2.5         4.9         1.5 versicolor
## 74          6.1         2.8         4.7         1.2 versicolor
## 75          6.4         2.9         4.3         1.3 versicolor
## 76          6.6         3.0         4.4         1.4 versicolor
## 77          6.8         2.8         4.8         1.4 versicolor
```

```
## 78          6.7         3.0         5.0         1.7 versicolor
## 79          6.0         2.9         4.5         1.5 versicolor
## 80          5.7         2.6         3.5         1.0 versicolor
## 81          5.5         2.4         3.8         1.1 versicolor
## 82          5.5         2.4         3.7         1.0 versicolor
## 83          5.8         2.7         3.9         1.2 versicolor
## 84          6.0         2.7         5.1         1.6 versicolor
## 85          5.4         3.0         4.5         1.5 versicolor
## 86          6.0         3.4         4.5         1.6 versicolor
## 87          6.7         3.1         4.7         1.5 versicolor
## 88          6.3         2.3         4.4         1.3 versicolor
## 89          5.6         3.0         4.1         1.3 versicolor
## 90          5.5         2.5         4.0         1.3 versicolor
## 91          5.5         2.6         4.4         1.2 versicolor
## 92          6.1         3.0         4.6         1.4 versicolor
## 93          5.8         2.6         4.0         1.2 versicolor
## 94          5.0         2.3         3.3         1.0 versicolor
## 95          5.6         2.7         4.2         1.3 versicolor
## 96          5.7         3.0         4.2         1.2 versicolor
## 97          5.7         2.9         4.2         1.3 versicolor
## 98          6.2         2.9         4.3         1.3 versicolor
## 99          5.1         2.5         3.0         1.1 versicolor
## 100         5.7         2.8         4.1         1.3 versicolor
## 101         6.3         3.3         6.0         2.5  virginica
## 102         5.8         2.7         5.1         1.9  virginica
## 103         7.1         3.0         5.9         2.1  virginica
## 104         6.3         2.9         5.6         1.8  virginica
## 105         6.5         3.0         5.8         2.2  virginica
```

```
## 106          7.6          3.0          6.6          2.1  virginica
## 107          4.9          2.5          4.5          1.7  virginica
## 108          7.3          2.9          6.3          1.8  virginica
## 109          6.7          2.5          5.8          1.8  virginica
## 110          7.2          3.6          6.1          2.5  virginica
## 111          6.5          3.2          5.1          2.0  virginica
## 112          6.4          2.7          5.3          1.9  virginica
## 113          6.8          3.0          5.5          2.1  virginica
## 114          5.7          2.5          5.0          2.0  virginica
## 115          5.8          2.8          5.1          2.4  virginica
## 116          6.4          3.2          5.3          2.3  virginica
## 117          6.5          3.0          5.5          1.8  virginica
## 118          7.7          3.8          6.7          2.2  virginica
## 119          7.7          2.6          6.9          2.3  virginica
## 120          6.0          2.2          5.0          1.5  virginica
## 121          6.9          3.2          5.7          2.3  virginica
## 122          5.6          2.8          4.9          2.0  virginica
## 123          7.7          2.8          6.7          2.0  virginica
## 124          6.3          2.7          4.9          1.8  virginica
## 125          6.7          3.3          5.7          2.1  virginica
## 126          7.2          3.2          6.0          1.8  virginica
## 127          6.2          2.8          4.8          1.8  virginica
## 128          6.1          3.0          4.9          1.8  virginica
## 129          6.4          2.8          5.6          2.1  virginica
## 130          7.2          3.0          5.8          1.6  virginica
## 131          7.4          2.8          6.1          1.9  virginica
## 132          7.9          3.8          6.4          2.0  virginica
## 133          6.4          2.8          5.6          2.2  virginica
```

```
## 134          6.3         2.8          5.1         1.5  virginica
## 135          6.1         2.6          5.6         1.4  virginica
## 136          7.7         3.0          6.1         2.3  virginica
## 137          6.3         3.4          5.6         2.4  virginica
## 138          6.4         3.1          5.5         1.8  virginica
## 139          6.0         3.0          4.8         1.8  virginica
## 140          6.9         3.1          5.4         2.1  virginica
## 141          6.7         3.1          5.6         2.4  virginica
## 142          6.9         3.1          5.1         2.3  virginica
## 143          5.8         2.7          5.1         1.9  virginica
## 144          6.8         3.2          5.9         2.3  virginica
## 145          6.7         3.3          5.7         2.5  virginica
## 146          6.7         3.0          5.2         2.3  virginica
## 147          6.3         2.5          5.0         1.9  virginica
## 148          6.5         3.0          5.2         2.0  virginica
## 149          6.2         3.4          5.4         2.3  virginica
## 150          5.9         3.0          5.1         1.8  virginica
```

```r
summary(iris)
```

```
##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width   
##  Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100  
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300  
##  Median :5.800   Median :3.000   Median :4.350   Median :1.300  
##  Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199  
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800  
##  Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500  
##        Species  
##  setosa    :50  
```

```
##  versicolor:50
##  virginica :50
##
##
##
```

```r
class(iris)
```

```
## [1] "data.frame"
```

```r
typeof(iris)
```

```
## [1] "list"
```

```r
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
## 2          4.9         3.0          1.4         0.2  setosa
## 3          4.7         3.2          1.3         0.2  setosa
## 4          4.6         3.1          1.5         0.2  setosa
## 5          5.0         3.6          1.4         0.2  setosa
## 6          5.4         3.9          1.7         0.4  setosa
```

```r
tail(iris)
```

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
## 145          6.7         3.3          5.7         2.5 virginica
```

```
## 146          6.7         3.0          5.2         2.3 virginica
## 147          6.3         2.5          5.0         1.9 virginica
## 148          6.5         3.0          5.2         2.0 virginica
## 149          6.2         3.4          5.4         2.3 virginica
## 150          5.9         3.0          5.1         1.8 virginica
```

```r
str(iris)
```

```
## 'data.frame':    150 obs. of  5 variables:
##  $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
##  $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
##  $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
##  $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
##  $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1
```

```r
species_habitat = data.frame(Species = c("setosa", "versicolor",
↪  "virginica"), Habitat = c("Woodland", "Meadow", "Mountain"))
species_habitat
```

```
##      Species  Habitat
## 1     setosa Woodland
## 2 versicolor   Meadow
## 3  virginica Mountain
```

```r
merged_dataset = merge(iris, species_habitat, by = "Species")
merged_dataset
```

```
##        Species Sepal.Length Sepal.Width Petal.Length Petal.Width  Habitat
## 1       setosa          5.1         3.5          1.4         0.2 Woodland
```

```
## 2        setosa          4.9          3.0          1.4          0.2 Woodland
## 3        setosa          4.7          3.2          1.3          0.2 Woodland
## 4        setosa          4.6          3.1          1.5          0.2 Woodland
## 5        setosa          5.0          3.6          1.4          0.2 Woodland
## 6        setosa          5.4          3.9          1.7          0.4 Woodland
## 7        setosa          4.6          3.4          1.4          0.3 Woodland
## 8        setosa          5.0          3.4          1.5          0.2 Woodland
## 9        setosa          4.4          2.9          1.4          0.2 Woodland
## 10       setosa          4.9          3.1          1.5          0.1 Woodland
## 11       setosa          5.4          3.7          1.5          0.2 Woodland
## 12       setosa          4.8          3.4          1.6          0.2 Woodland
## 13       setosa          4.8          3.0          1.4          0.1 Woodland
## 14       setosa          4.3          3.0          1.1          0.1 Woodland
## 15       setosa          5.8          4.0          1.2          0.2 Woodland
## 16       setosa          5.7          4.4          1.5          0.4 Woodland
## 17       setosa          5.4          3.9          1.3          0.4 Woodland
## 18       setosa          5.1          3.5          1.4          0.3 Woodland
## 19       setosa          5.7          3.8          1.7          0.3 Woodland
## 20       setosa          5.1          3.8          1.5          0.3 Woodland
## 21       setosa          5.4          3.4          1.7          0.2 Woodland
## 22       setosa          5.1          3.7          1.5          0.4 Woodland
## 23       setosa          4.6          3.6          1.0          0.2 Woodland
## 24       setosa          5.1          3.3          1.7          0.5 Woodland
## 25       setosa          4.8          3.4          1.9          0.2 Woodland
## 26       setosa          5.0          3.0          1.6          0.2 Woodland
## 27       setosa          5.0          3.4          1.6          0.4 Woodland
## 28       setosa          5.2          3.5          1.5          0.2 Woodland
## 29       setosa          5.2          3.4          1.4          0.2 Woodland
```

```
## 30      setosa          4.7         3.2          1.6         0.2 Woodland
## 31      setosa          4.8         3.1          1.6         0.2 Woodland
## 32      setosa          5.4         3.4          1.5         0.4 Woodland
## 33      setosa          5.2         4.1          1.5         0.1 Woodland
## 34      setosa          5.5         4.2          1.4         0.2 Woodland
## 35      setosa          4.9         3.1          1.5         0.2 Woodland
## 36      setosa          5.0         3.2          1.2         0.2 Woodland
## 37      setosa          5.5         3.5          1.3         0.2 Woodland
## 38      setosa          4.9         3.6          1.4         0.1 Woodland
## 39      setosa          4.4         3.0          1.3         0.2 Woodland
## 40      setosa          5.1         3.4          1.5         0.2 Woodland
## 41      setosa          5.0         3.5          1.3         0.3 Woodland
## 42      setosa          4.5         2.3          1.3         0.3 Woodland
## 43      setosa          4.4         3.2          1.3         0.2 Woodland
## 44      setosa          5.0         3.5          1.6         0.6 Woodland
## 45      setosa          5.1         3.8          1.9         0.4 Woodland
## 46      setosa          4.8         3.0          1.4         0.3 Woodland
## 47      setosa          5.1         3.8          1.6         0.2 Woodland
## 48      setosa          4.6         3.2          1.4         0.2 Woodland
## 49      setosa          5.3         3.7          1.5         0.2 Woodland
## 50      setosa          5.0         3.3          1.4         0.2 Woodland
## 51  versicolor          7.0         3.2          4.7         1.4   Meadow
## 52  versicolor          6.4         3.2          4.5         1.5   Meadow
## 53  versicolor          6.9         3.1          4.9         1.5   Meadow
## 54  versicolor          5.5         2.3          4.0         1.3   Meadow
## 55  versicolor          6.5         2.8          4.6         1.5   Meadow
## 56  versicolor          5.7         2.8          4.5         1.3   Meadow
## 57  versicolor          6.3         3.3          4.7         1.6   Meadow
```

```
## 58  versicolor          4.9          2.4          3.3          1.0   Meadow
## 59  versicolor          6.6          2.9          4.6          1.3   Meadow
## 60  versicolor          5.2          2.7          3.9          1.4   Meadow
## 61  versicolor          5.0          2.0          3.5          1.0   Meadow
## 62  versicolor          5.9          3.0          4.2          1.5   Meadow
## 63  versicolor          6.0          2.2          4.0          1.0   Meadow
## 64  versicolor          6.1          2.9          4.7          1.4   Meadow
## 65  versicolor          5.6          2.9          3.6          1.3   Meadow
## 66  versicolor          6.7          3.1          4.4          1.4   Meadow
## 67  versicolor          5.6          3.0          4.5          1.5   Meadow
## 68  versicolor          5.8          2.7          4.1          1.0   Meadow
## 69  versicolor          6.2          2.2          4.5          1.5   Meadow
## 70  versicolor          5.6          2.5          3.9          1.1   Meadow
## 71  versicolor          5.9          3.2          4.8          1.8   Meadow
## 72  versicolor          6.1          2.8          4.0          1.3   Meadow
## 73  versicolor          6.3          2.5          4.9          1.5   Meadow
## 74  versicolor          6.1          2.8          4.7          1.2   Meadow
## 75  versicolor          6.4          2.9          4.3          1.3   Meadow
## 76  versicolor          6.6          3.0          4.4          1.4   Meadow
## 77  versicolor          6.8          2.8          4.8          1.4   Meadow
## 78  versicolor          6.7          3.0          5.0          1.7   Meadow
## 79  versicolor          6.0          2.9          4.5          1.5   Meadow
## 80  versicolor          5.7          2.6          3.5          1.0   Meadow
## 81  versicolor          5.5          2.4          3.8          1.1   Meadow
## 82  versicolor          5.5          2.4          3.7          1.0   Meadow
## 83  versicolor          5.8          2.7          3.9          1.2   Meadow
## 84  versicolor          6.0          2.7          5.1          1.6   Meadow
## 85  versicolor          5.4          3.0          4.5          1.5   Meadow
```

```
## 86  versicolor          6.0          3.4          4.5          1.6   Meadow
## 87  versicolor          6.7          3.1          4.7          1.5   Meadow
## 88  versicolor          6.3          2.3          4.4          1.3   Meadow
## 89  versicolor          5.6          3.0          4.1          1.3   Meadow
## 90  versicolor          5.5          2.5          4.0          1.3   Meadow
## 91  versicolor          5.5          2.6          4.4          1.2   Meadow
## 92  versicolor          6.1          3.0          4.6          1.4   Meadow
## 93  versicolor          5.8          2.6          4.0          1.2   Meadow
## 94  versicolor          5.0          2.3          3.3          1.0   Meadow
## 95  versicolor          5.6          2.7          4.2          1.3   Meadow
## 96  versicolor          5.7          3.0          4.2          1.2   Meadow
## 97  versicolor          5.7          2.9          4.2          1.3   Meadow
## 98  versicolor          6.2          2.9          4.3          1.3   Meadow
## 99  versicolor          5.1          2.5          3.0          1.1   Meadow
## 100 versicolor          5.7          2.8          4.1          1.3   Meadow
## 101  virginica          6.3          3.3          6.0          2.5 Mountain
## 102  virginica          5.8          2.7          5.1          1.9 Mountain
## 103  virginica          7.1          3.0          5.9          2.1 Mountain
## 104  virginica          6.3          2.9          5.6          1.8 Mountain
## 105  virginica          6.5          3.0          5.8          2.2 Mountain
## 106  virginica          7.6          3.0          6.6          2.1 Mountain
## 107  virginica          4.9          2.5          4.5          1.7 Mountain
## 108  virginica          7.3          2.9          6.3          1.8 Mountain
## 109  virginica          6.7          2.5          5.8          1.8 Mountain
## 110  virginica          7.2          3.6          6.1          2.5 Mountain
## 111  virginica          6.5          3.2          5.1          2.0 Mountain
## 112  virginica          6.4          2.7          5.3          1.9 Mountain
## 113  virginica          6.8          3.0          5.5          2.1 Mountain
```

```
## 114  virginica          5.7          2.5          5.0          2.0 Mountain
## 115  virginica          5.8          2.8          5.1          2.4 Mountain
## 116  virginica          6.4          3.2          5.3          2.3 Mountain
## 117  virginica          6.5          3.0          5.5          1.8 Mountain
## 118  virginica          7.7          3.8          6.7          2.2 Mountain
## 119  virginica          7.7          2.6          6.9          2.3 Mountain
## 120  virginica          6.0          2.2          5.0          1.5 Mountain
## 121  virginica          6.9          3.2          5.7          2.3 Mountain
## 122  virginica          5.6          2.8          4.9          2.0 Mountain
## 123  virginica          7.7          2.8          6.7          2.0 Mountain
## 124  virginica          6.3          2.7          4.9          1.8 Mountain
## 125  virginica          6.7          3.3          5.7          2.1 Mountain
## 126  virginica          7.2          3.2          6.0          1.8 Mountain
## 127  virginica          6.2          2.8          4.8          1.8 Mountain
## 128  virginica          6.1          3.0          4.9          1.8 Mountain
## 129  virginica          6.4          2.8          5.6          2.1 Mountain
## 130  virginica          7.2          3.0          5.8          1.6 Mountain
## 131  virginica          7.4          2.8          6.1          1.9 Mountain
## 132  virginica          7.9          3.8          6.4          2.0 Mountain
## 133  virginica          6.4          2.8          5.6          2.2 Mountain
## 134  virginica          6.3          2.8          5.1          1.5 Mountain
## 135  virginica          6.1          2.6          5.6          1.4 Mountain
## 136  virginica          7.7          3.0          6.1          2.3 Mountain
## 137  virginica          6.3          3.4          5.6          2.4 Mountain
## 138  virginica          6.4          3.1          5.5          1.8 Mountain
## 139  virginica          6.0          3.0          4.8          1.8 Mountain
## 140  virginica          6.9          3.1          5.4          2.1 Mountain
## 141  virginica          6.7          3.1          5.6          2.4 Mountain
```

```
## 142  virginica          6.9          3.1           5.1          2.3 Mountain
## 143  virginica          5.8          2.7           5.1          1.9 Mountain
## 144  virginica          6.8          3.2           5.9          2.3 Mountain
## 145  virginica          6.7          3.3           5.7          2.5 Mountain
## 146  virginica          6.7          3.0           5.2          2.3 Mountain
## 147  virginica          6.3          2.5           5.0          1.9 Mountain
## 148  virginica          6.5          3.0           5.2          2.0 Mountain
## 149  virginica          6.2          3.4           5.4          2.3 Mountain
## 150  virginica          5.9          3.0           5.1          1.8 Mountain
```

Q4

Create a two-dimensional 5×3 array of sequence of even integers > 50

```
v = seq(52, 80, 2)
a = array(v, dim = c(5,3))
a
```

```
##      [,1] [,2] [,3]
## [1,]   52   62   72
## [2,]   54   64   74
## [3,]   56   66   76
## [4,]   58   68   78
## [5,]   60   70   80
```

Q5

Create a vector which contains 10 integer values between -50 and +50

```
vec = seq(-45, 45, 10)
vec
```

```
##  [1] -45 -35 -25 -15  -5   5  15  25  35  45
```

Q6

Suppose age is a vector containing ages of 10 persons. Solve the given questions.

```r
#(a) Access the age of fourth person
age = c(22, 27, 31, 41, 30, 25, 19, 20, 23, 35)
age
```

```
##  [1] 22 27 31 41 30 25 19 20 23 35
```

```r
age[4]
```

```
## [1] 41
```

```r
#(b) Vector with age>30
age30 = age[age > 30]
age30
```

```
## [1] 31 41 35
```

```r
#(c) Ages of last 3 persons
age[8:10]
```

```
## [1] 20 23 35
```

```r
#(d) No of elements in vector age
length(age)
```

```
## [1] 10
```

```r
#(e) Ages of persons except 5th and 7th
age[c(-5, -7)]
```

```
## [1] 22 27 31 41 25 20 23 35
```

```r
#(f) Vector with ages between 20 and 25
age2 = age[age > 20 & age < 25]
age2
```

```
## [1] 22 23
```

Q7

Create factor from given data and find levels

```r
f = factor(c(1,2,3,2,3,1,4,2,3,NA,5,3,2))
f
```

```
##  [1] 1    2    3    2    3    1    4    2    3    <NA> 5    3    2
## Levels: 1 2 3 4 5
```

```r
levels(f)
```

```
## [1] "1" "2" "3" "4" "5"
```

Q8

List containing strings, numbers, vectors and logical values

```
l = list(15L, 2.5, 9.3, 10L, "probability", FALSE, "Statistics",
 ↪  c(1,2,3), seq(1, 10, 3), TRUE)
l
```

```
## [[1]]
## [1] 15
##
## [[2]]
## [1] 2.5
##
## [[3]]
## [1] 9.3
##
## [[4]]
## [1] 10
##
## [[5]]
## [1] "probability"
##
## [[6]]
## [1] FALSE
##
## [[7]]
## [1] "Statistics"
##
```

```
## [[8]]
## [1] 1 2 3
##
## [[9]]
## [1]  1  4  7 10
##
## [[10]]
## [1] TRUE
```

Q9

Using built-in dataset iris, find out categorical variables

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
names(select_if(iris, is.factor))
```

```
## [1] "Species"
```

Q10

Create a numeric vector c(1:5) and a 5 by 3 matrix with elements from 1 to 15

```r
n = c(1:5)
n
```

```
## [1] 1 2 3 4 5
```

Q11

Create a dataframe of the given dataset and solve the questions

```r
df = data.frame(height = c(140,137,150,147,139,140,150,132,138,140),
 ↪  weight = c(55,57,59,62,61,60,60,58,59,57))
df #Required dataframe
```

```
##    height weight
## 1     140     55
```

```
## 2      137      57
## 3      150      59
## 4      147      62
## 5      139      61
## 6      140      60
## 7      150      60
## 8      132      58
## 9      138      59
## 10     140      57
```

```r
#(a) Vector h1 with height > 145 cm
h1 = df$height[df$height>145]
h1
```

```
## [1] 150 147 150
```

```r
#(b) Vector h2 with weight > 55 kg
h2 = df$height[df$weight>55]
h2
```

```
## [1] 137 150 147 139 140 150 132 138 140
```

```r
#(c) Vector h3 with height > 140 and weight > 60
h3 = df$height[df$height>140 & df$weight>60]
h3
```

```
## [1] 147
```